Lan Pan

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RESEARCH INTERESTS

Computational cognitive science; human and machine decision-making; AI for science; behavioral data infrastructure; language-based measurement; LLM behavior and cognitive modeling.

EDUCATION

Southwestern University of Finance and Economics, China
Double Bachelor's Degree in Marketing and Finance, Sep 2015 - Jun 2019
GPA: 3.5, Rank: 6/105

Undergraduate theses:

- Marketing: The factors influencing purchase intention for new energy vehicles.
- Finance: An empirical study on the relationship between RMB exchange-rate fluctuations and the international balance of payments.

PUBLICATIONS

1. **Pan, L.***, Xie, H.*, & Wilson, R. C. (2025). Large language models think too fast to explore effectively. *Advances in Neural Information Processing Systems, 38*, 102480-102508. https://doi.org/10.52202/085713-3428
2. **Pan, L.**, & Musslick, S. (2027). Has someone already tested this? Construct-level hypothesis-literature matching. In M. Cerrato, J. N. van Rijn, & S. Kramer (Eds.), *Discovery Science* (DS 2026, Lecture Notes in Computer Science, Vol. 17130, pp. 234-248). Springer. https://doi.org/10.1007/978-3-032-41086-3_16
3. Xie, H., Jagadish, A. K., **Pan, L.**, & Wilson, R. C. (2026). *Think-aloud reshapes automated cognitive model discovery beyond behavior* [Conference presentation]. 9th Annual Conference on Cognitive Computational Neuroscience (CCN). https://arxiv.org/abs/2605.05091

* Equal contribution.

MANUSCRIPTS AND WORKING PAPERS

1. "Behavioral-Computational Task Space of Cognitive Tasks." Manuscript in preparation.

RESEARCH EXPERIENCE

1. **Research Assistant, NRD Lab, Georgia Institute of Technology**
   Supervisor: Robert C. Wilson; Hanbo Xie
   Nov 2024 - Present

- Conduct research on large language models as computational models of human learning and decision-making.
- **Project 1 (NeurIPS 2025):** Led research evaluating whether large language models can explore effectively in an open-ended discovery task, benchmarked against large-scale human behavior. Identified systematic gaps in exploratory performance across model families and characterized their decision patterns using a compact strategy analysis.
- **Project 2 (ongoing):** Investigate cross-task generalization of LLMs trained on individual psychology tasks, analyzing similarities and differences across task-induced behaviors.

2. **Research Collaborator**
   Collaborators: Akshay K. Jagadish and Younes Strittmatter, Princeton University
   Jun 2026 - Present

- **Ongoing:** Develop an agent that answers researchers' queries by identifying relevant evidence across a cross-task collection of thousands of behavioral datasets and analyzing the selected datasets.

3. **Research Assistant, Automated Scientific Discovery of Mind and Brain Lab, University of Osnabrueck**
   Supervisor: Sebastian Musslick
   Dec 2025 - Present

- Work on automated scientific discovery systems for cognitive science.
- **Project 1 (Discovery Science 2026):** Developed a construct-level hypothesis-literature matching pipeline that retrieves papers, extracts scientific hypotheses, and assesses construct-level matches.
- **Project 2 (ongoing):** Develop a label-free event-state representation of cognitive-control paradigms to compare task structures across studies and test when differently named paradigms may rely on shared mechanisms.

PROFESSIONAL EXPERIENCE

1. **Chengdu Jiaotou Huaizhou Xincheng Investment and Operation Co., Ltd.**, China
   Investment and Operation
   Nov 2021 - Jun 2023
2. **Chengdu Youzhujialian Group Co., Ltd.**, China
   Planning Management Trainee; Planning Manager
   Jul 2019 - Nov 2021

ACADEMIC SERVICE

Reviewer, Frontiers of Computer Science, 2025

SKILLS

- Programming and data analysis: Python, pandas, NumPy, statistical modeling, machine learning workflows.
- Behavioral and cognitive modeling: regression, ANOVA, PCA, factor analysis, task-performance analysis, cross-task transfer analysis.
- LLM and NLP methods: LoRA fine-tuning, embeddings, semantic similarity, natural language inference, model deployment, quantization.
- Scientific infrastructure: reproducible analysis pipelines, dataset integrity checks, provenance tracking, agentic analysis workflows.
- Writing and tools: LaTeX, Git.

SELECTED COURSEWORK

Advanced Mathematics; Linear Algebra; Probability Theory; Statistics; Econometrics; Consumer Behavior; Market Research; Database Management Systems; Macroeconomics; Microeconomics; Corporate Finance; Investments; International Finance; Derivative Financial Instruments.